

# Customer Relationship Management (CRM)

## Using Agile Methodology

### 1. Abstract

Customer Relationship Management (CRM) is a software system that helps businesses manage interactions with current and potential customers. In a competitive market, maintaining strong customer relationships is the key to business growth. This project focuses on developing a CRM system using the Agile Software Development Methodology. Unlike traditional models, Agile allows for the continuous delivery of useful software modules, such as lead tracking, contact management, and sales reporting, in short cycles called sprints. Traditional CRM development often faces challenges like rigid requirements and delayed feedback. Agile methodology solves this by breaking the project into manageable sprints. Each sprint delivers a specific feature—for example, the first sprint might focus on "User Management," while the second focuses on "Lead Capture." This ensures that the development team can adapt to changes quickly and that the client gets to see working software early in the process.

The CRM system allows Admin users to assign tasks, Sales Agents to update lead statuses, and Managers to view performance reports. By using Agile ceremonies like Sprint Planning and Daily Stand-ups, the team ensures better collaboration and transparency. The final product streamlines sales processes, centralizes customer data, and improves customer retention rates. This project demonstrates the effectiveness of Agile in building scalable, user-centric business applications.

### 2. Introduction

#### 2.1 Introduction

A CRM system is designed to streamline sales, marketing, and customer support. This project utilizes Agile methodology to build a flexible and efficient CRM solution.

#### 2.2 Problem Identification

Businesses often struggle with scattered customer data, missed follow-ups, and lack of visibility into the sales pipeline, leading to lost revenue.

### 2.3 Need of the Project

A centralized platform is needed to organize customer information, track sales progress, and automate communication.

### 2.4 Project Scheduling

| Phase         | Duration |
|---------------|----------|
| Planning      | 2 Days   |
| Development   | 8 Days   |
| Testing       | 3 Days   |
| Review        | 2 Days   |
| Documentation | 1 Day    |

### 2.5 Objectives

- Centralize customer data storage.
- Automate lead tracking and follow-ups.
- Implement Agile development practices.
- Provide real-time sales analytics.
- Enhance team collaboration.

### **3. Software Requirement Specification (SRS)**

#### **3.1 Purpose**

To develop a dynamic CRM application that improves business relationships and sales efficiency using Agile.

#### **3.2 Scope**

Useful for sales teams, marketing departments, customer support agents, and business managers.

#### **3.3 Hardware / Software Requirement**

##### **Hardware:**

- 4GB RAM
- Intel i3 or above
- 500GB HDD

##### **Software:**

- Windows 10 / 11
- Java / Web Technologies
- MySQL Database
- VS Code / Eclipse IDE

#### **3.4 Tools**

- HTML, CSS, JavaScript (Frontend)
- Java / Spring Boot (Backend)
- MySQL (Database)
- Git (Version Control)
- Jira / Trello (Agile Project Management)

### 3.5 Software Process Model

- **Agile Model**
- Sprint Planning
- Daily Stand-up Meetings
- Sprint Review
- Sprint Retrospective

## 4. System Design

### 4.1 Data Dictionary

| Field      | Type   | Description                    |
|------------|--------|--------------------------------|
| CustomerID | int    | Unique Customer ID             |
| Name       | String | Customer Full Name             |
| Email      | String | Contact Email                  |
| Phone      | String | Contact Number                 |
| LeadStatus | String | New/Contacted/Qualified/Closed |
| AssignedTo | int    | ID of Sales Agent              |

## 4.2 ER Diagram

Admin → Sales Agent → Customer → Interaction History

## 4.3 DFD

Customer Data → CRM System → Database → Sales Report

## 4.4 Diagrams

- Use Case Diagram (Actors: Admin, Agent)
- Activity Diagram (Lead Conversion Process)
- Flowchart

## 5. Implementation

### 5.1 Program Code

Modules developed in sprints:

- **Sprint 1:** User Authentication (Login/Role Management)
- **Sprint 2:** Contact Management (Add/Edit/Delete Customers)
- **Sprint 3:** Lead Tracking (Status Updates)
- **Sprint 4:** Task Scheduler (Reminders)
- **Sprint 5:** Reporting Dashboard

### 5.2 Output Screens

- Admin Dashboard
- Add New Lead Form
- Customer List View
- Sales Performance Chart
- Task Reminder Popup

## 6. Testing

### 6.1 Test Data

| Input                     | Expected Output                |
|---------------------------|--------------------------------|
| Valid Customer Details    | Customer Added Successfully    |
| Duplicate Email           | Error: Customer Already Exists |
| Update Status to 'Closed' | Sales count increments         |

### 6.2 Test Result

Unit testing and integration testing were performed after each sprint. All critical bugs were resolved during the sprint cycle.

## 7. User Manual

### 7.1 How to Use

1. **Login** with Agent credentials.
2. Click **"Add Lead"** to enter new customer details.
3. Go to **"My Leads"** to update status (e.g., change from 'New' to 'Contacted').
4. Use **"Tasks"** to set follow-up reminders.
5. View **"Reports"** to see monthly sales.

### 7.2 Screen Layout

A clean, responsive dashboard with a side navigation bar for easy access to modules.

## **8. Project Applications & Limitations**

### **8.1 Applications**

- Sales Pipeline Management
- Customer Support Ticketing
- Marketing Campaign Tracking
- Client Data Organization

### **8.2 Limitations**

- Requires manual data entry for initial leads.
- Training required for non-technical staff.
- Dependency on internet connectivity.

## **9. Conclusion & Future Enhancement**

### **Conclusion:**

The CRM system developed using Agile methodology provides a robust solution for managing business relationships. The iterative nature of Agile allowed the team to incorporate feedback and ensure high usability.

### **Future Enhancements:**

- Integration with Email (Gmail/Outlook)
- AI-based lead scoring
- Mobile App for field sales agents
- Voice command features

## **10. Bibliography & References**

- Agile Manifesto
- CRM Best Practices Guide
- Java Documentation
- Software Engineering by Roger Pressman
- Stack Overflow & GeeksForGeeks